Phase 2 rubric

What we'll check

Area	What we expect
Infrastructure and GitHub	Use Databricks (Free Edition is fine). All your PySpark code goes in GitHub with a sensible folder structure.
Data models and README	Document every Bronze and Silver table in your README: column names, data types and the primary key.
Schemas and data types	Define the schema with StructType/Field before reading any file. Don't use inferSchema. Cast every field to its proper type when it moves to Silver.
load_timestamp	Every row in every Bronze and Silver table needs a load_timestamp showing when it was loaded.
Bronze layer	Load the raw data as it arrives and keep track of where each record came from (source, file or batch).
Silver layer and data quality	Clean and validate the data, fix the types, and deal with duplicates and bad records.
MERGE and idempotency	Use MERGE INTO to insert new records and update existing ones. If you run the same input twice, Silver should look exactly the same afterwards, with no duplicate rows.
Parameters and backfills	Both steps, Raw to Bronze and Bronze to Silver, should take a parameter such as a date, batch ID or folder path. You should be able to re-run any past period without editing the code.
Schema drift	If the source adds a column or changes a data type, the pipeline must not crash. Either evolve the schema (mergeSchema) or move the bad records to a quarantine table and let the rest of the batch finish.
Execution logs	Keep a log table, for example pipeline_execution_logs. Every run writes a row to it, for full and incremental loads, in both steps. Each row should have the layer, the file or parameter processed, start and end time, status (success or failure), and the number of rows inserted and updated.
Code and run guide	Keep the code readable. Your README should explain how to run a normal incremental load and how to run a backfill for an older period.

What to submit

- Your GitHub repository link.
- The notebooks or scripts for Raw to Bronze and Bronze to Silver.
- A README with your Bronze and Silver data models and the run instructions (incremental load and backfill).
- Proof that it works. Either leave the notebook outputs saved in the repo, or record a short screen video (around 5 minutes). Show the explicit schemas, a MERGE re-run that creates no duplicates, a backfill, how a schema change is handled, and your log table.

Please note

Grading will be done on what your pipeline does, not what the README says it does. If a feature only exists in a comment or a plan, it won't get marks. Make sure we can run or see every feature in your environment.